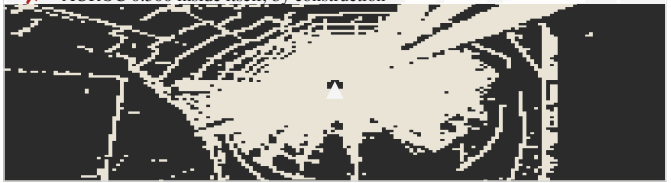
the input

six surround cameras. no LiDAR, no map, no HD prior.
the pedestrian is behind the post, into direct sun.



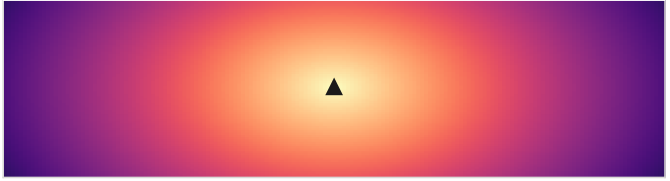
mask_camera the dataset's own flag

- ✗ binary. a voxel is in or out, nothing between
- ✗ built offline from the full sensor suite, including the LiDAR the model never receives
- ✗ not available at inference
- ✗ AUROC 0.500 inside itself, by construction



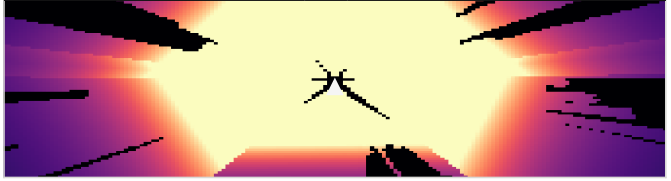
range from ego

- ✓ continuous, and needs no network
- ✗ ignores occlusion entirely: a brick wall and open road at the same distance score alike
- ✗ AUROC 0.666 per object



observability ours

- ✓ continuous, and carries one parameter
- ✓ camera-only: known before the model runs
- ✓ resolves occlusion by first-hit ray marching
- ✓ AUROC 0.762 per object, +0.0484 over mask_camera
- ✓ within noise of a LiDAR oracle (0.774)



All four panels are the same frame, token dc03fe4e0a37, scene scene-0553, sealed test split. AUROC values are over 40,863 objects in 75 scenes.